

Class05: Data Vis with ggplot2

Qiman Hu A17912409

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Background

There are many graphics systems in R for making plots and figures. These include so-called “*base R*” *graphics* like the `plot()` function and add on packages like **ggplot2**.

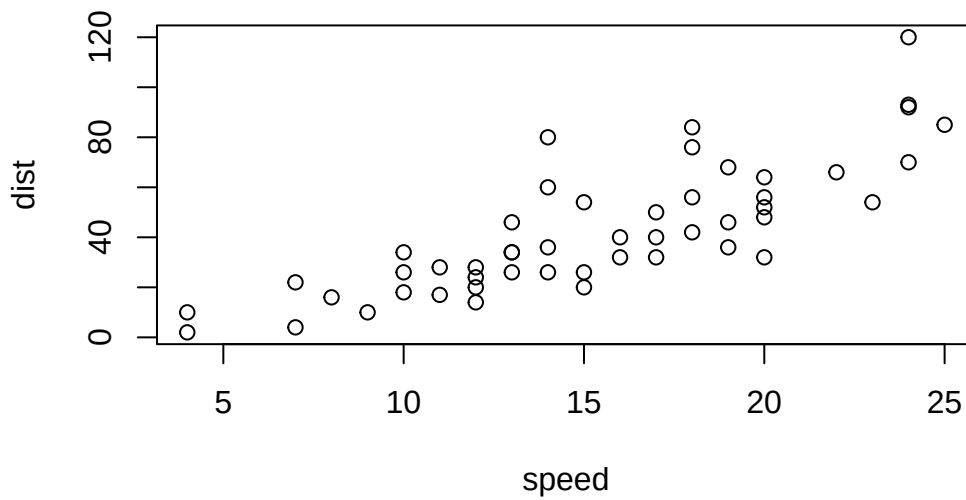
Let’s compare how we make a simple figure with these two systems:

We can use the in-built `cars` dataset::

```
head (cars)
```

```
  speed dist
1     4    2
2     4   10
3     7    4
4     7   22
5     8   16
6     9   10
```

```
plot(cars)
```



Before I can use **ggplot2** I need to install it on my computer. To do this we can use the function `install.package("ggplot2")`

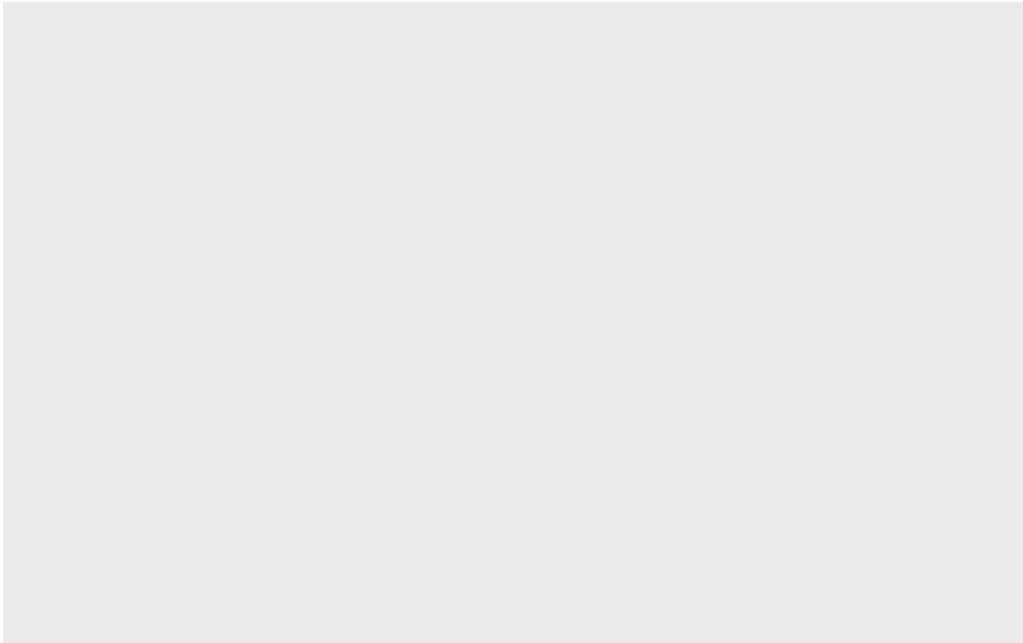
N.B. We never use `install.package("ggplot2")` in our quarto doc (we run it once only in our R console) as it would re-install the package every time we render our report.

Once installed we need to load up the package into our R brain:

```
library(ggplot2)
```

The main function in the **ggplot2** package is called `ggplot()`

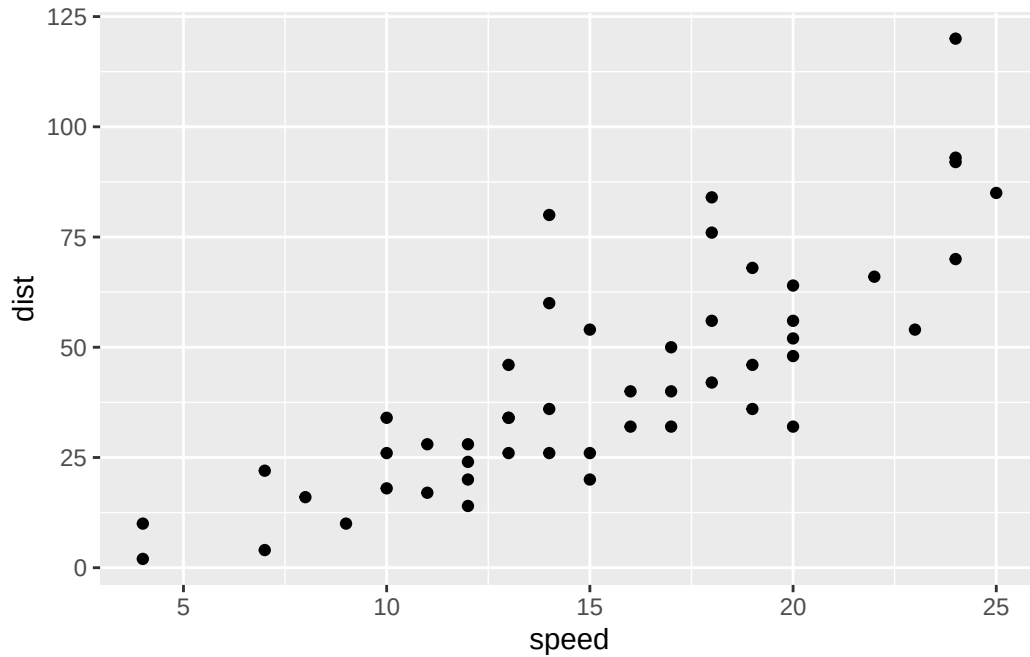
```
ggplot(cars)
```



Every ggplot has at least 3 layers:

- the **data** (a data.frame of the stuff we want to plot)
- the **aesthetics** (how the data maps to the plot)
- the **geom** layer (how you want the plot drawn, e.g. points, lines, etc.)

```
ggplot(cars) +  
  aes(x= speed , y= dist) +  
  geom_point()
```

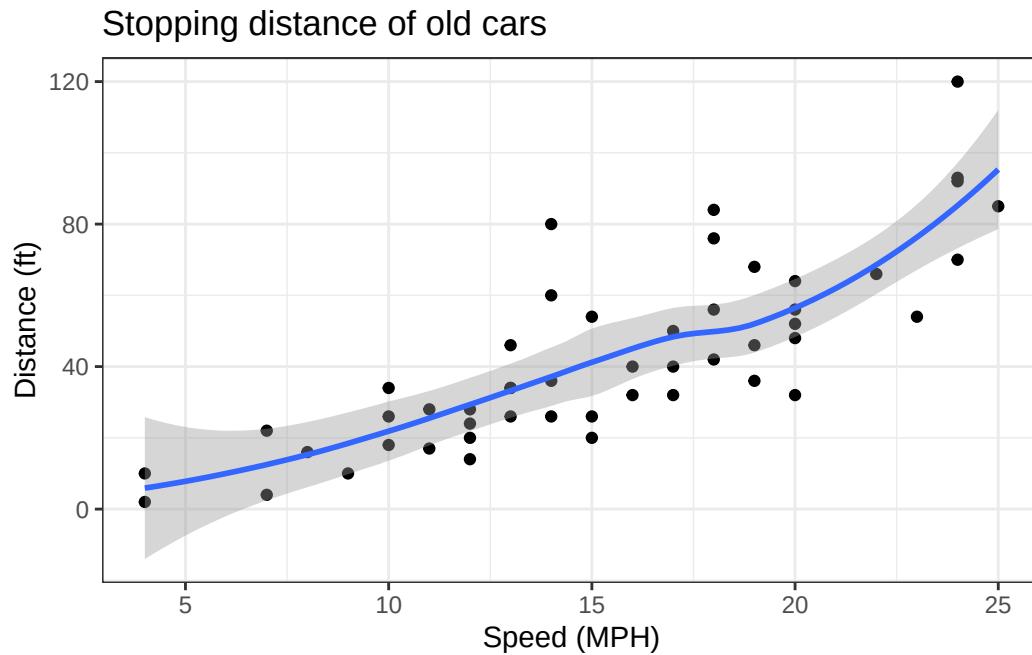


Add some custom features

Let's add a trend line that shows the relationship between speed and distance.

```
ggplot(cars) +  
  aes(x= speed , y= dist) +  
  geom_point() +  
  geom_smooth()+  
  theme_bw()+  
  labs(title= "Stopping distance of old cars", x= "Speed (MPH)", y="Distance (ft)")
```

``geom_smooth()`` using `method = 'loess'` and `formula = 'y ~ x'`

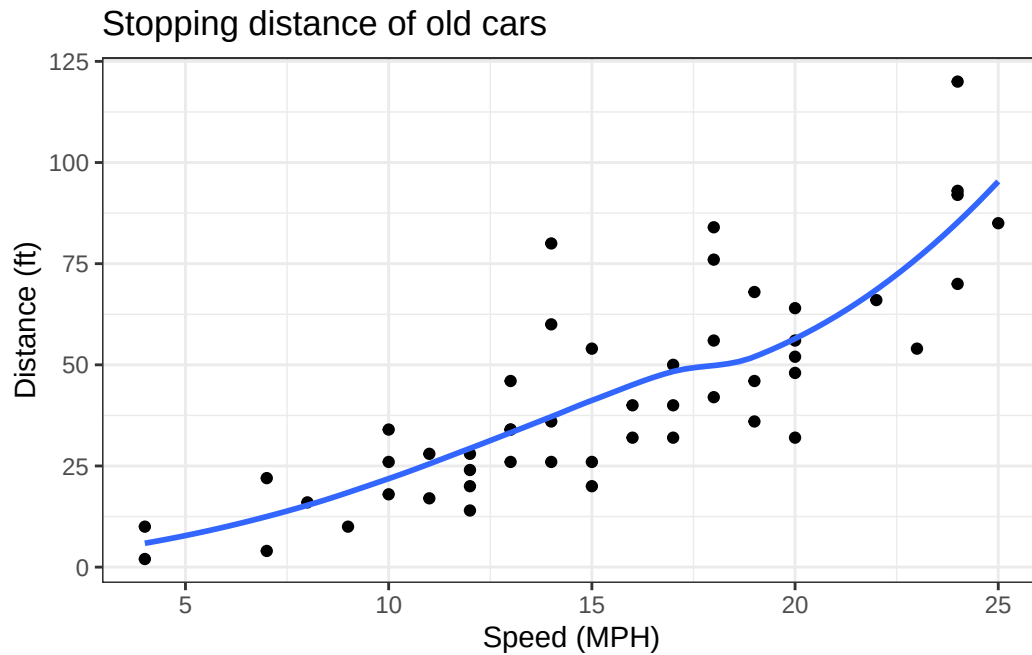


Q. Can you make the `geom_smooth()` function produce a linear straight line fit to the data and turn off the “gray” error region?

```
ggplot(cars) +
  aes(x= speed , y= dist) +
  geom_point() +
  geom_smooth(methods= "lm", se= FALSE)+
  theme_bw()+
  labs(title= "Stopping distance of old cars", x= "Speed (MPH)", y="Distance (ft)")
```

Warning in `geom_smooth(methods = "lm", se = FALSE)`: Ignoring unknown parameters: `methods`

`geom\_smooth()` using method = 'loess' and formula = 'y ~ x'



Gene Expression

Import the data from course website

```
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"
genes <- read.delim(url)
head(genes)
```

	Gene	Condition1	Condition2	State
1	A4GNT	-3.6808610	-3.4401355	unchanging
2	AAAS	4.5479580	4.3864126	unchanging
3	AASDH	3.7190695	3.4787276	unchanging
4	AATF	5.0784720	5.0151916	unchanging
5	AATK	0.4711421	0.5598642	unchanging
6	AB015752.4	-3.6808610	-3.5921390	unchanging

```
sum(genes$State == "up")
```

```
[1] 127
```

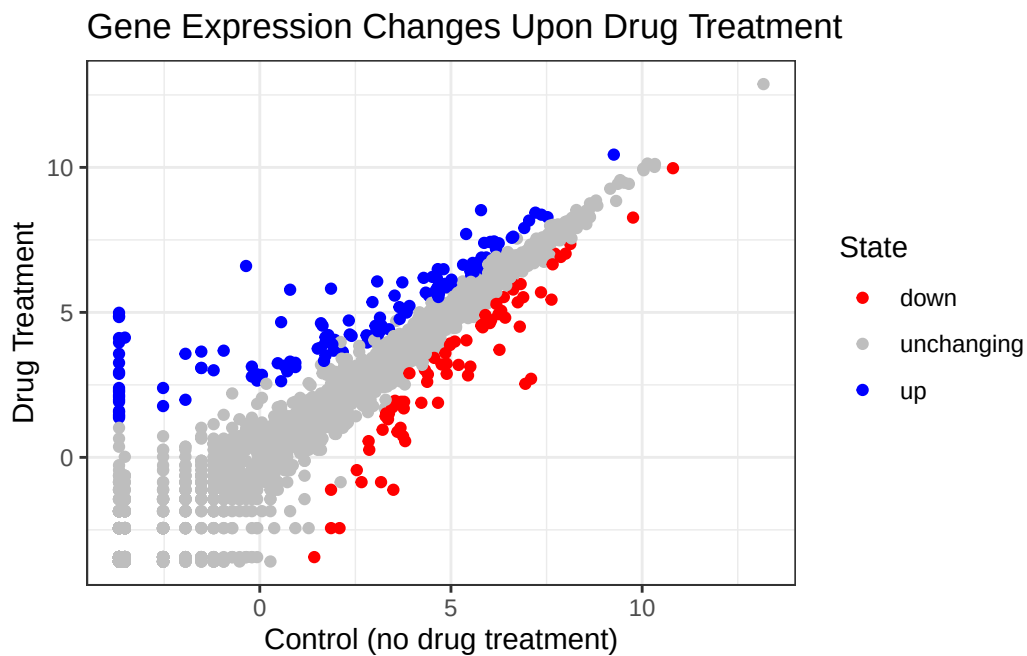
A useful new function in this context is the `table()` function:

```
table(genes$State)
```

down	unchanging	up
72	4997	127

My first plot attempt

```
ggplot(genes) +  
  aes(x= Condition1, y= Condition2, col=State)+  
  geom_point() +  
  labs(title = "Gene Expression Changes Upon Drug Treatment", x= "Control (no drug treatment.",  
  scale_color_manual(values= c("Red","gray", "Blue"))+  
  theme_bw()
```



```
table(genes$State)
```

down	unchanging	up
72	4997	127

Going further

here we read the famous gapminder dataset:

```
# File location online
url <- "https://raw.githubusercontent.com/jennybc/gapminder/master/inst/extdata/gapminder.tsv"

gapminder <- read.delim(url)

head(gapminder)
```

	country	continent	year	lifeExp	pop	gdpPercap
1	Afghanistan	Asia	1952	28.801	8425333	779.4453
2	Afghanistan	Asia	1957	30.332	9240934	820.8530
3	Afghanistan	Asia	1962	31.997	10267083	853.1007
4	Afghanistan	Asia	1967	34.020	11537966	836.1971
5	Afghanistan	Asia	1972	36.088	13079460	739.9811
6	Afghanistan	Asia	1977	38.438	14880372	786.1134

Q. How many entries (rows) are in this dataset?

```
nrow(gapminder)
```

```
[1] 1704
```

Q. how many “country” entries are there?

```
table(gapminder$country)
```

Afghanistan	Albania	Algeria
12	12	12
Angola	Argentina	Australia
12	12	12
Austria	Bahrain	Bangladesh
12	12	12
Belgium	Benin	Bolivia
12	12	12
Bosnia and Herzegovina	Botswana	Brazil
12	12	12
Bulgaria	Burkina Faso	Burundi

12	12	12
Cambodia	Cameroon	Canada
12	12	12
Central African Republic	Chad	Chile
12	12	12
China	Colombia	Comoros
12	12	12
Congo, Dem. Rep.	Congo, Rep.	Costa Rica
12	12	12
Cote d'Ivoire	Croatia	Cuba
12	12	12
Czech Republic	Denmark	Djibouti
12	12	12
Dominican Republic	Ecuador	Egypt
12	12	12
El Salvador	Equatorial Guinea	Eritrea
12	12	12
Ethiopia	Finland	France
12	12	12
Gabon	Gambia	Germany
12	12	12
Ghana	Greece	Guatemala
12	12	12
Guinea	Guinea-Bissau	Haiti
12	12	12
Honduras	Hong Kong, China	Hungary
12	12	12
Iceland	India	Indonesia
12	12	12
Iran	Iraq	Ireland
12	12	12
Israel	Italy	Jamaica
12	12	12
Japan	Jordan	Kenya
12	12	12
Korea, Dem. Rep.	Korea, Rep.	Kuwait
12	12	12
Lebanon	Lesotho	Liberia
12	12	12
Libya	Madagascar	Malawi
12	12	12
Malaysia	Mali	Mauritania
12	12	12

Mauritius	Mexico	Mongolia
12	12	12
Montenegro	Morocco	Mozambique
12	12	12
Myanmar	Namibia	Nepal
12	12	12
Netherlands	New Zealand	Nicaragua
12	12	12
Niger	Nigeria	Norway
12	12	12
Oman	Pakistan	Panama
12	12	12
Paraguay	Peru	Philippines
12	12	12
Poland	Portugal	Puerto Rico
12	12	12
Reunion	Romania	Rwanda
12	12	12
Sao Tome and Principe	Saudi Arabia	Senegal
12	12	12
Serbia	Sierra Leone	Singapore
12	12	12
Slovak Republic	Slovenia	Somalia
12	12	12
South Africa	Spain	Sri Lanka
12	12	12
Sudan	Swaziland	Sweden
12	12	12
Switzerland	Syria	Taiwan
12	12	12
Tanzania	Thailand	Togo
12	12	12
Trinidad and Tobago	Tunisia	Turkey
12	12	12
Uganda	United Kingdom	United States
12	12	12
Uruguay	Venezuela	Vietnam
12	12	12
West Bank and Gaza	Yemen, Rep.	Zambia
12	12	12
Zimbabwe		
12		

```
unique(gapminder$country)
```

[1] "Afghanistan"	"Albania"
[3] "Algeria"	"Angola"
[5] "Argentina"	"Australia"
[7] "Austria"	"Bahrain"
[9] "Bangladesh"	"Belgium"
[11] "Benin"	"Bolivia"
[13] "Bosnia and Herzegovina"	"Botswana"
[15] "Brazil"	"Bulgaria"
[17] "Burkina Faso"	"Burundi"
[19] "Cambodia"	"Cameroon"
[21] "Canada"	"Central African Republic"
[23] "Chad"	"Chile"
[25] "China"	"Colombia"
[27] "Comoros"	"Congo, Dem. Rep."
[29] "Congo, Rep."	"Costa Rica"
[31] "Cote d'Ivoire"	"Croatia"
[33] "Cuba"	"Czech Republic"
[35] "Denmark"	"Djibouti"
[37] "Dominican Republic"	"Ecuador"
[39] "Egypt"	"El Salvador"
[41] "Equatorial Guinea"	"Eritrea"
[43] "Ethiopia"	"Finland"
[45] "France"	"Gabon"
[47] "Gambia"	"Germany"
[49] "Ghana"	"Greece"
[51] "Guatemala"	"Guinea"
[53] "Guinea-Bissau"	"Haiti"
[55] "Honduras"	"Hong Kong, China"
[57] "Hungary"	"Iceland"
[59] "India"	"Indonesia"
[61] "Iran"	"Iraq"
[63] "Ireland"	"Israel"
[65] "Italy"	"Jamaica"
[67] "Japan"	"Jordan"
[69] "Kenya"	"Korea, Dem. Rep."
[71] "Korea, Rep."	"Kuwait"
[73] "Lebanon"	"Lesotho"
[75] "Liberia"	"Libya"
[77] "Madagascar"	"Malawi"
[79] "Malaysia"	"Mali"

[81]	"Mauritania"	"Mauritius"
[83]	"Mexico"	"Mongolia"
[85]	"Montenegro"	"Morocco"
[87]	"Mozambique"	"Myanmar"
[89]	"Namibia"	"Nepal"
[91]	"Netherlands"	"New Zealand"
[93]	"Nicaragua"	"Niger"
[95]	"Nigeria"	"Norway"
[97]	"Oman"	"Pakistan"
[99]	"Panama"	"Paraguay"
[101]	"Peru"	"Philippines"
[103]	"Poland"	"Portugal"
[105]	"Puerto Rico"	"Reunion"
[107]	"Romania"	"Rwanda"
[109]	"Sao Tome and Principe"	"Saudi Arabia"
[111]	"Senegal"	"Serbia"
[113]	"Sierra Leone"	"Singapore"
[115]	"Slovak Republic"	"Slovenia"
[117]	"Somalia"	"South Africa"
[119]	"Spain"	"Sri Lanka"
[121]	"Sudan"	"Swaziland"
[123]	"Sweden"	"Switzerland"
[125]	"Syria"	"Taiwan"
[127]	"Tanzania"	"Thailand"
[129]	"Togo"	"Trinidad and Tobago"
[131]	"Tunisia"	"Turkey"
[133]	"Uganda"	"United Kingdom"
[135]	"United States"	"Uruguay"
[137]	"Venezuela"	"Vietnam"
[139]	"West Bank and Gaza"	"Yemen, Rep."
[141]	"Zambia"	"Zimbabwe"

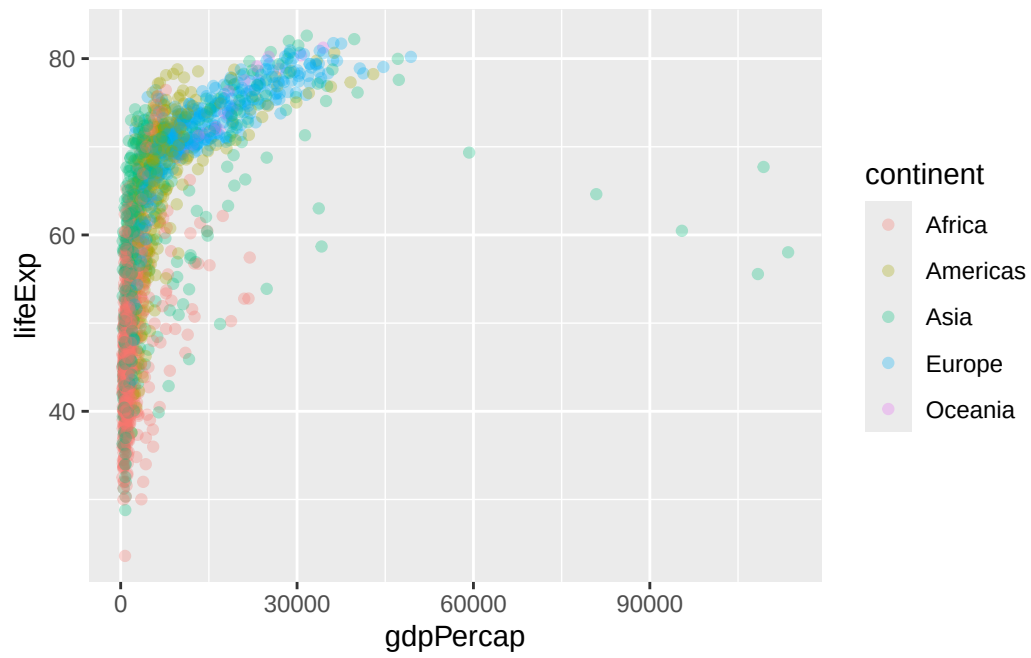
```
length(unique(gapminder$country))
```

```
[1] 142
```

Lets make our first plot of the entire dataset: Plot of “gdpPercap” VS “lifeExp“ colored by”continent”

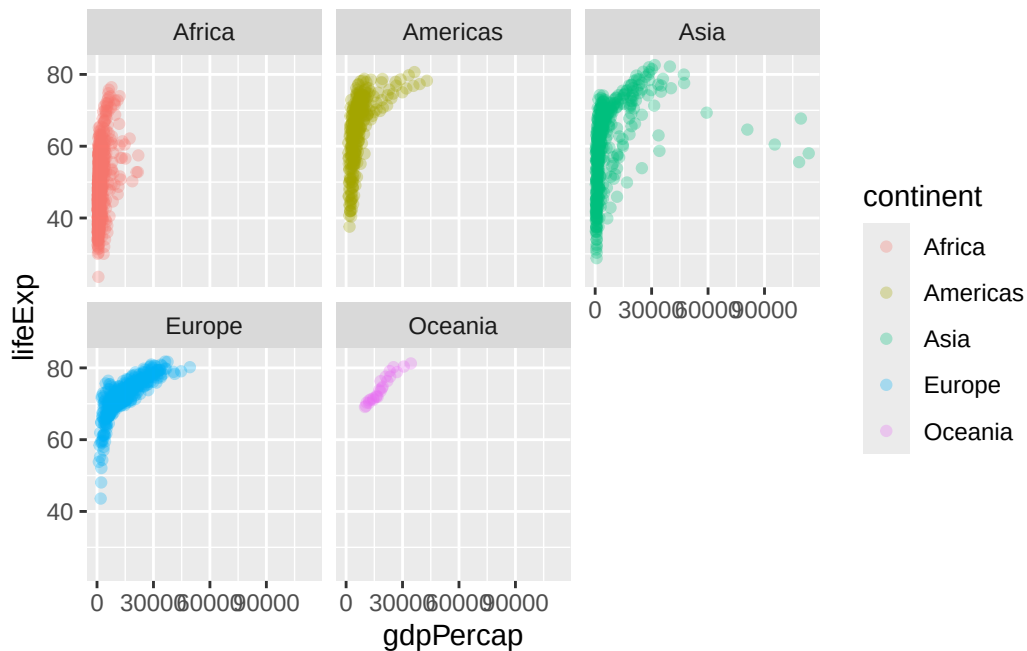
```
p <- ggplot(gapminder)+
aes(gdpPercap, lifeExp, col= continent )+
geom_point(alpha= 0.3)
```

p



p+

```
facet_wrap(~continent)
```



Make a plot for years 1977 and 2007 only.

Q. First use the **dplyr** package and the **filter()** function from that package to extract the rows from the year 2007

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

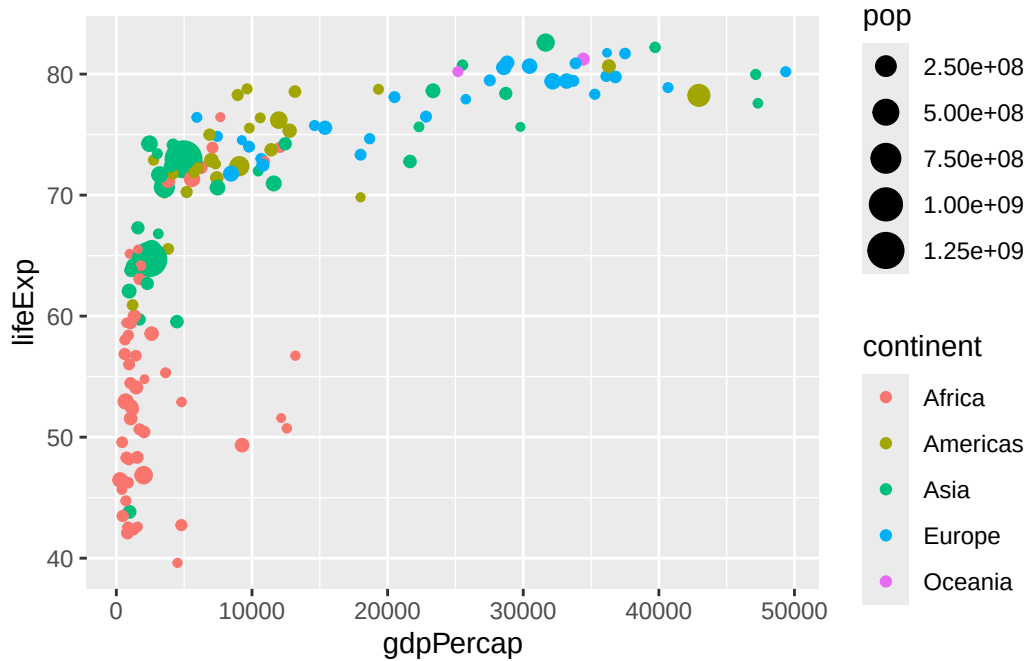
`filter`, `lag`

The following objects are masked from 'package:base':

`intersect`, `setdiff`, `setequal`, `union`

```
g07 <- gapminder %>% filter(year==2007)
```

```
ggplot(g07)+
  aes(gdpPercap, lifeExp, col=continent, size=pop) +
  geom_point( )
```



Do the same for 1977

```
g77 <- gapminder %>% filter(year==1977)
g <- filter(gapminder, year== 2007 | year== 1977)
```

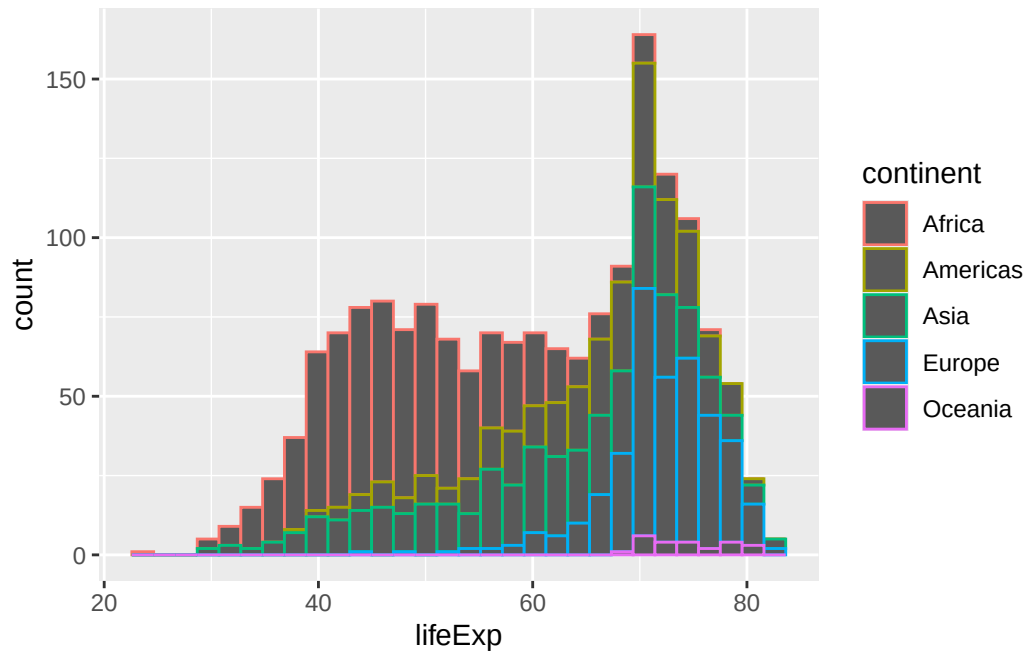
```
ggplot(g)+
  aes(gdpPercap, lifeExp, col=continent, size=pop) +
  geom_point( )+
  facet_wrap(~year)
```



Q. Make a histogram of life Exp colored by continent (try `fill=continent` or `col=continent`)

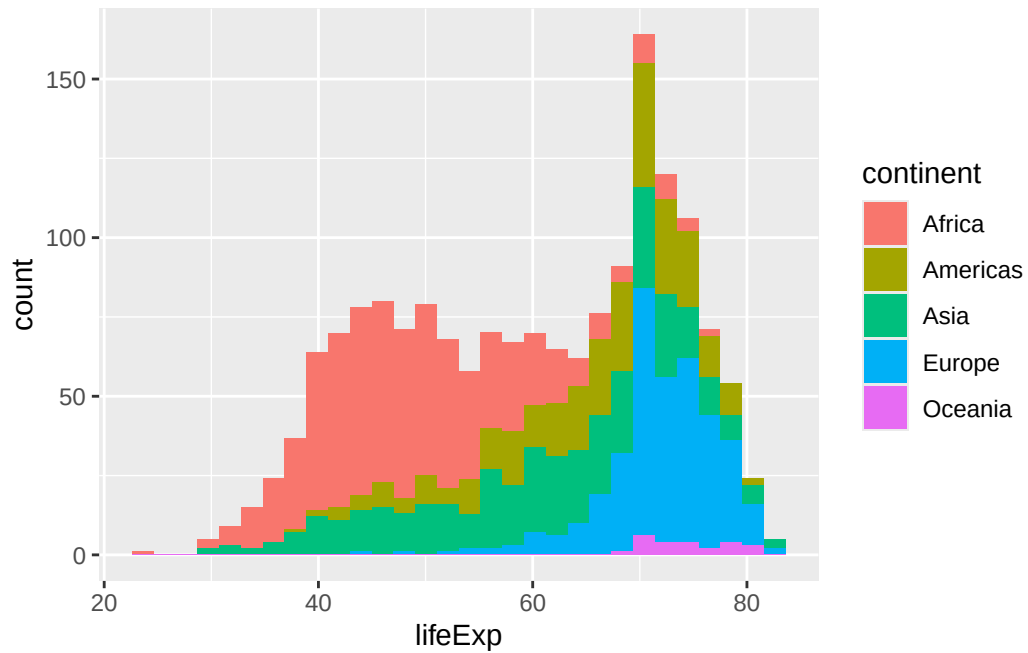
```
ggplot(gapminder)+
  aes(lifeExp, col=continent) +
  geom_histogram()
```

``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



```
ggplot(gapminder)+  
  aes(lifeExp, fill=continent) +  
  geom_histogram()
```

``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Q. Make a histogram of life Exp faceted by continent

```
ggplot(gapminder)+
  aes(lifeExp) +
  geom_histogram()+
  facet_wrap(~continent)
```

`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.

